

L9

Walking Gait

Walking



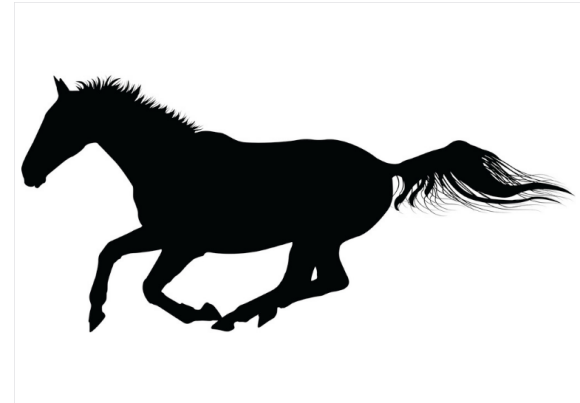
<https://newsroom.uhc.com/health/Move-More-Month.html>

Limping



<https://www.dreamstime.com/photos-images/limping.html>

Galloping



<https://www.vectorstock.com/royalty-free-vector/galloping-horse-silhouette-vector-776365>

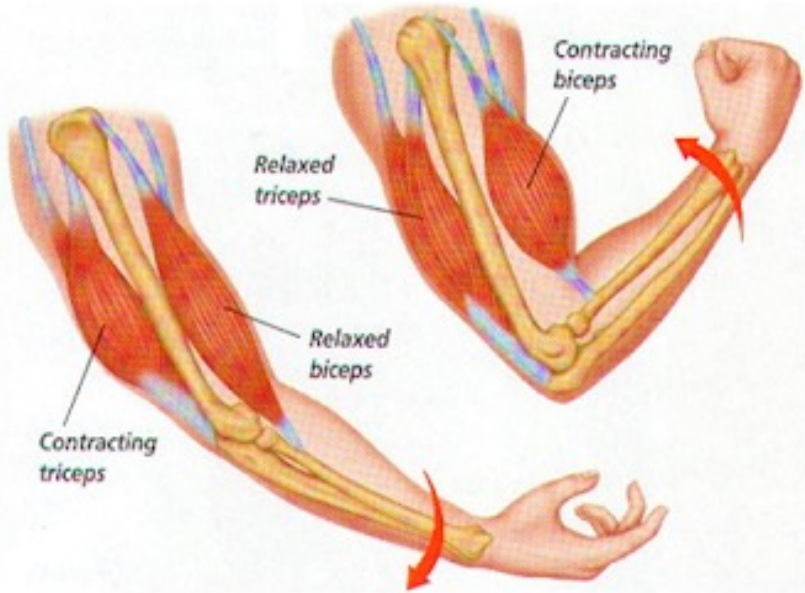
Trotting



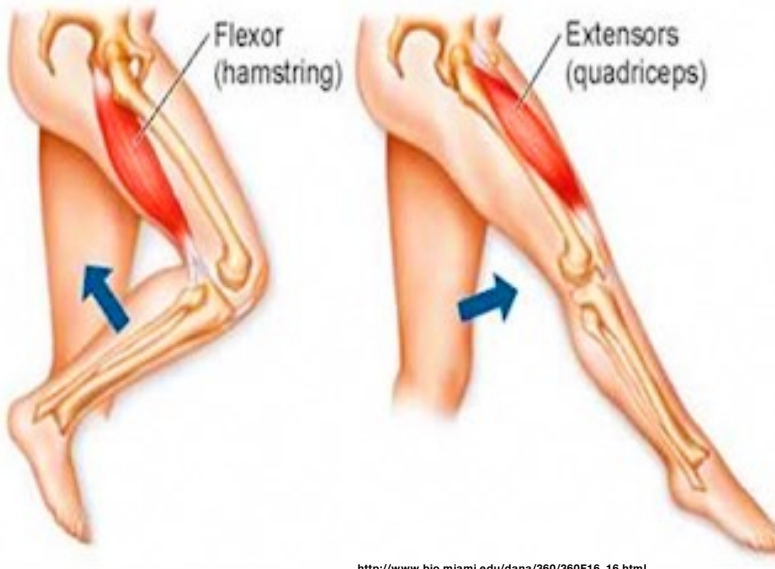
<https://en.wikipedia.org/wiki/Trot>

Way of walking

Principle of reciprocity



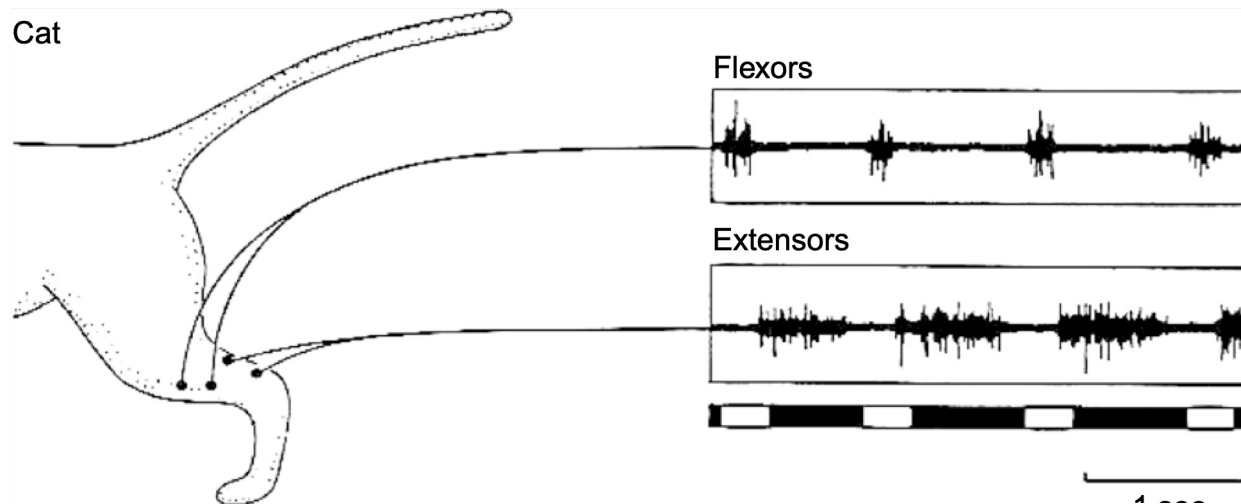
- Muscles tend to arrange in antagonist pairs
- To avoid counteracting each other function that could suppress movement



Sherrington law of reciprocal innervation

Muscular activity during walking

EMG Recording

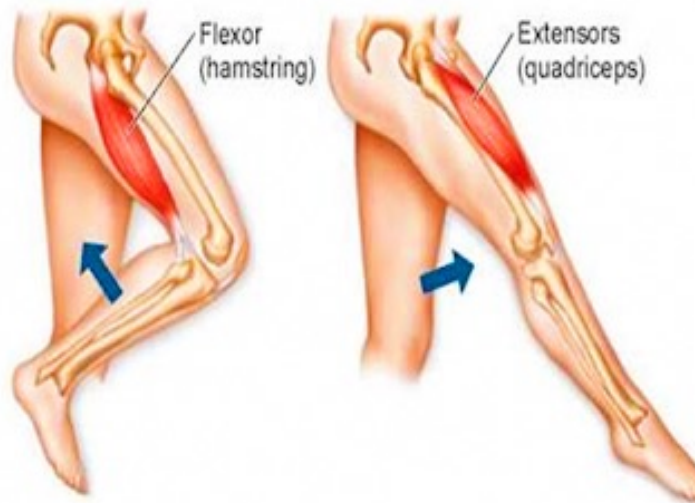


Swing phase

Flexor Active

Stance phase

Extensors Active

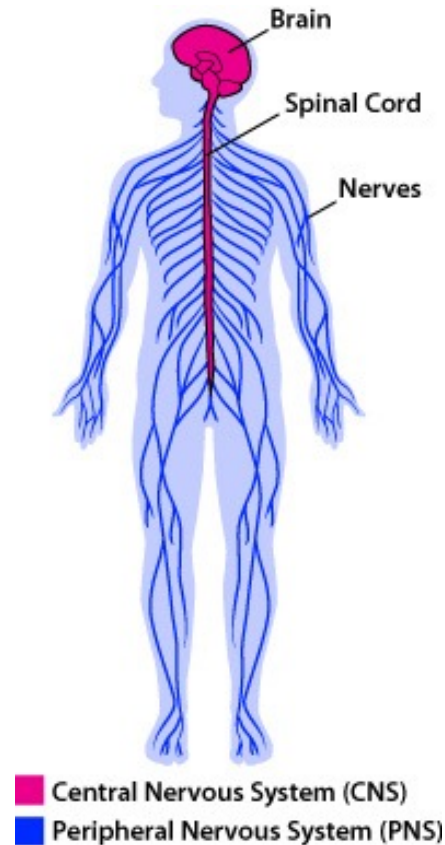


Brain & Spinal Cord

Charles Scott Sherrington, 1906

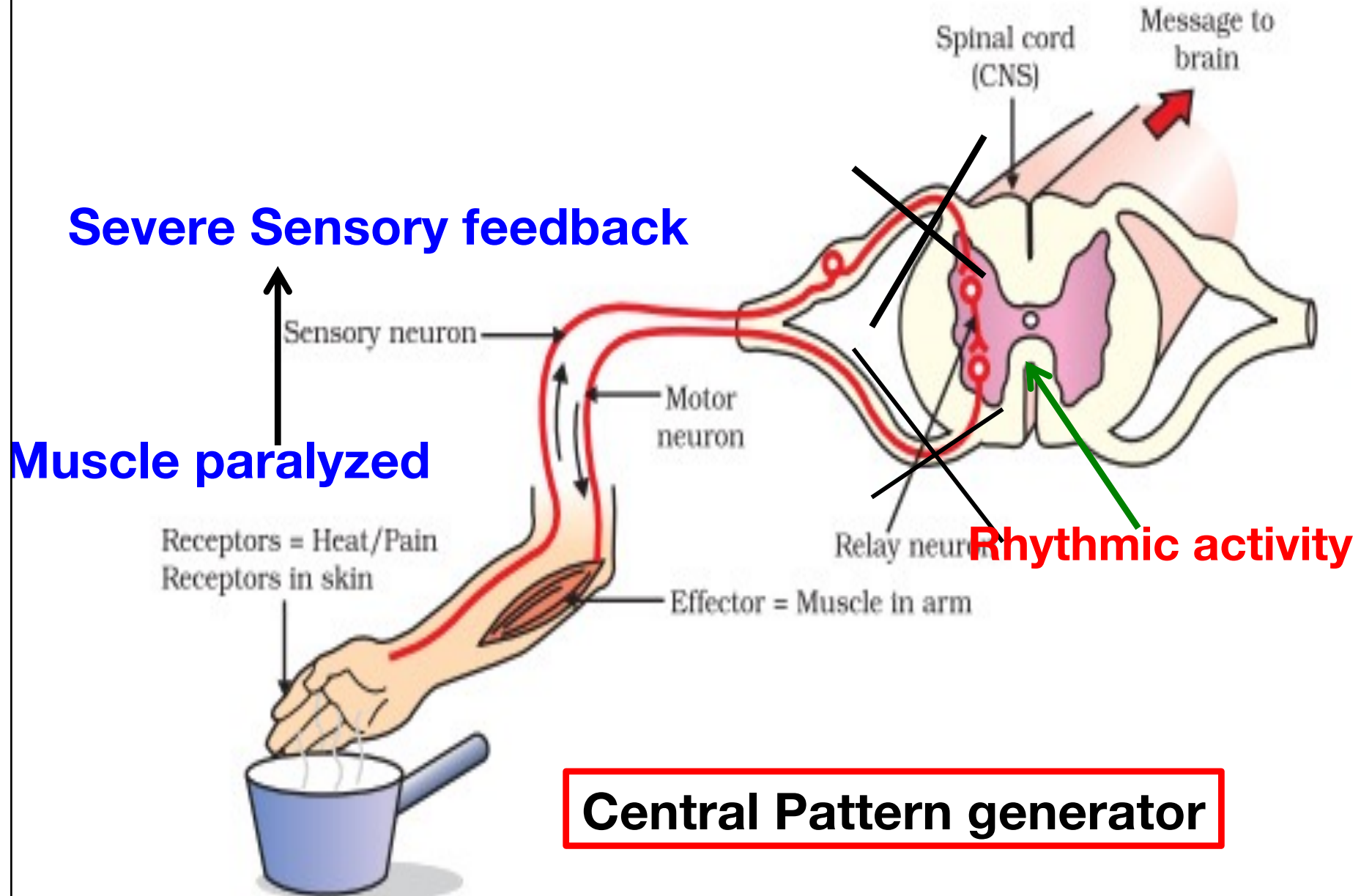


https://en.wikipedia.org/wiki/Charles_Scott_Sherrington



- Showed dissociation between brain and the spinal cord
- Spinal cord can produce rhythmic movement **without** input from the brain.

Fictive Locomotion

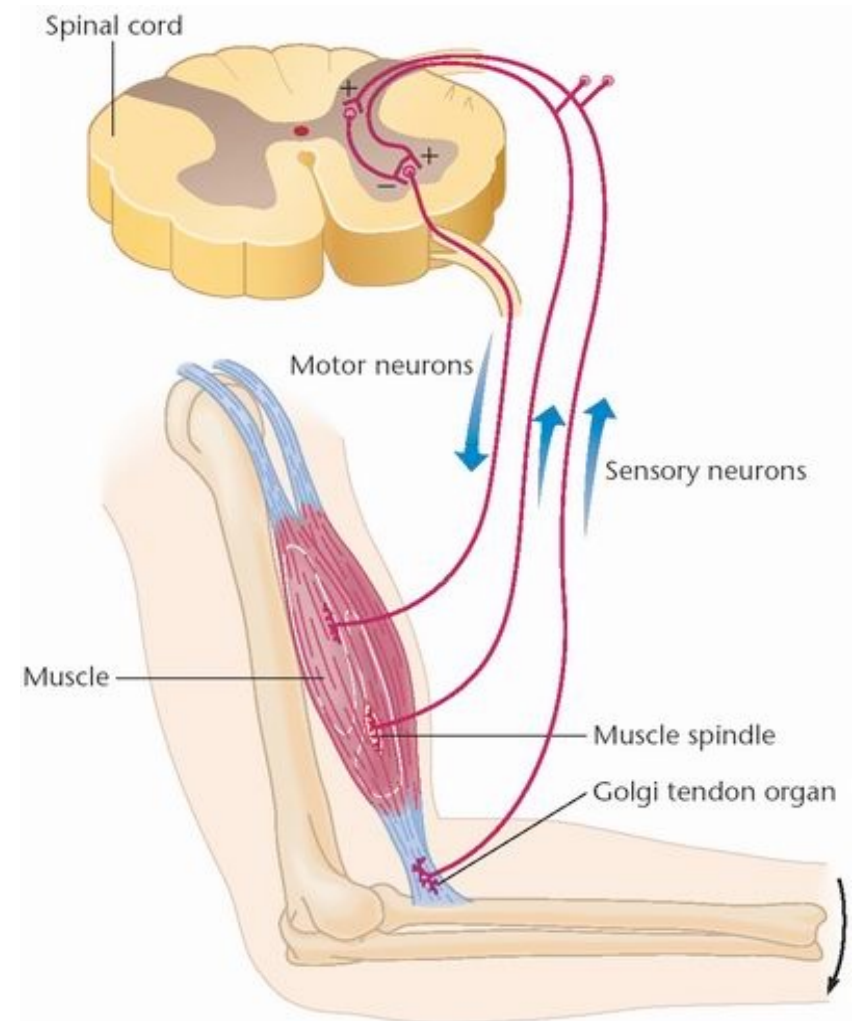


Proprioception

Perception of information from receptors present in the muscles, joints, tendons and skin.

Kinesthesia

- Muscle spindles
- Golgi tendon organs
- Joint receptors
- Cutaneous receptors

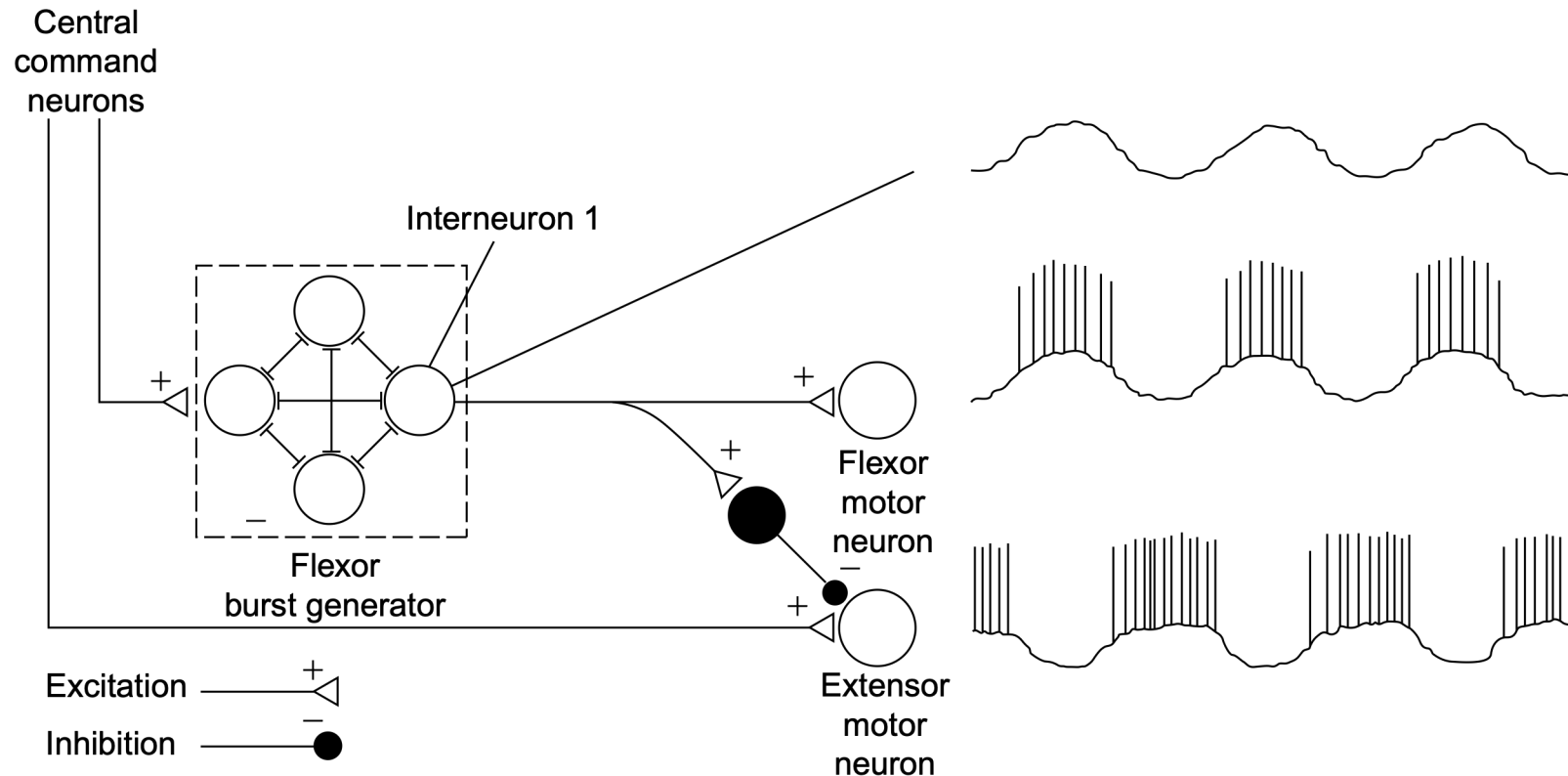


Spinal cord self sufficient in initiating movement

Role of Brain?

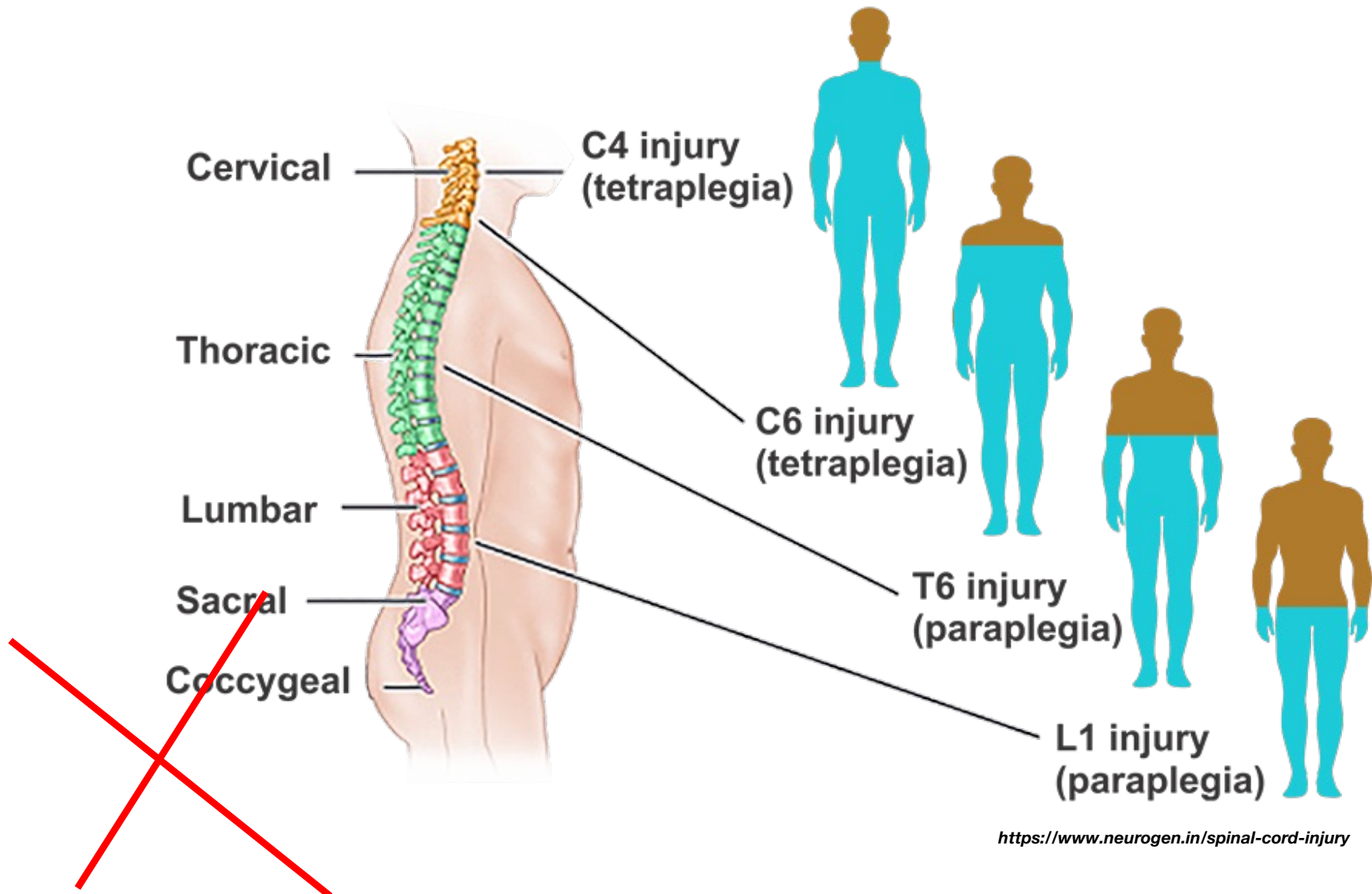
Neural circuits of locomotion

Central Pattern Generator (CPG)



**Neural circuit for cockroach locomotion
hypothesized by Pearson (1976)**

Spinal Cord Injuries

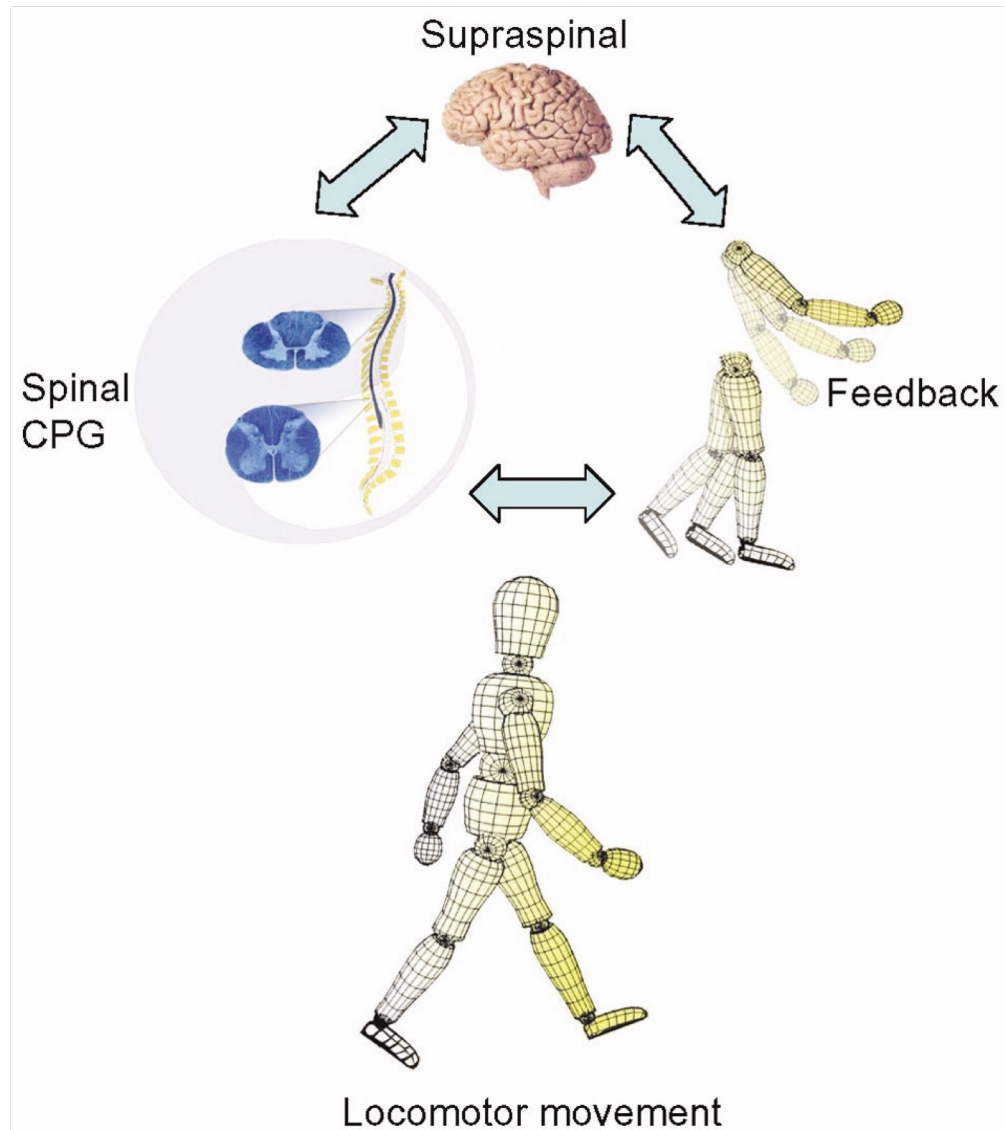


**Spinal cord self sufficient in
initiating movement**

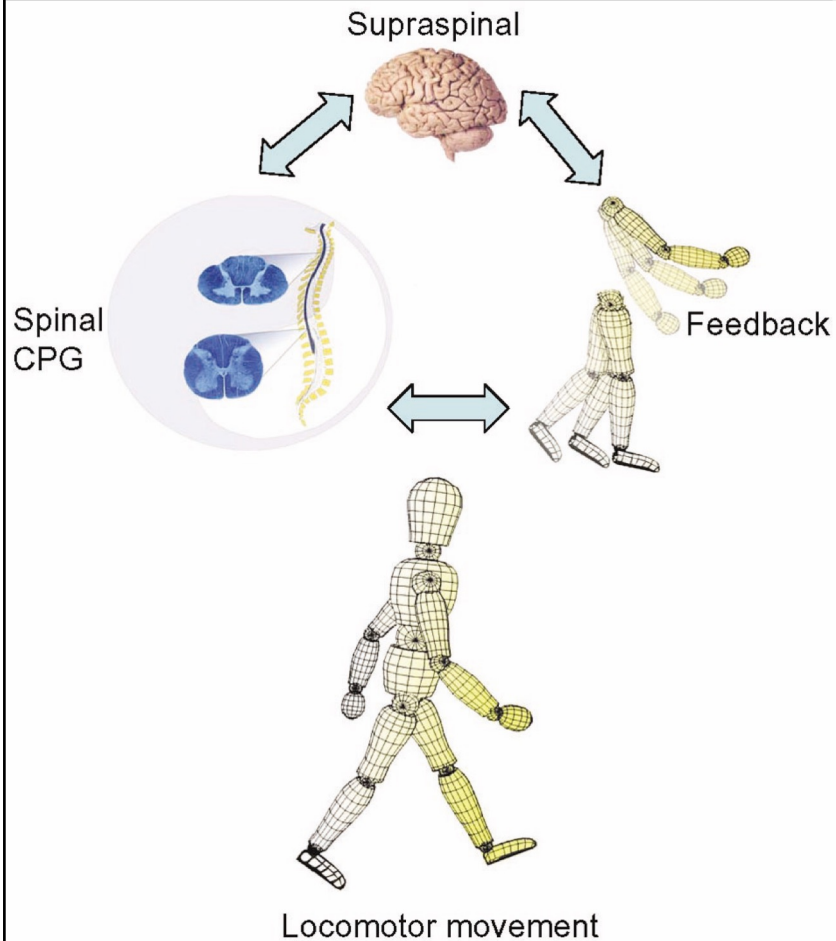
Role of Brain?

Neural circuits of locomotion

Neural circuits of locomotion

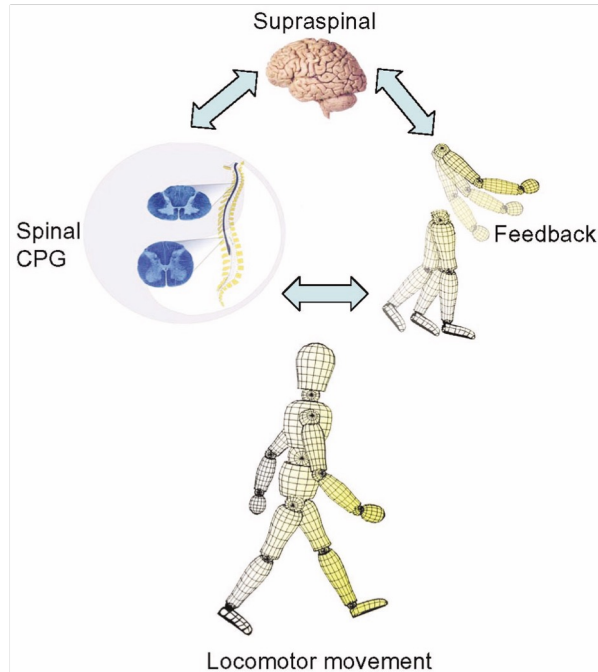


Sensory Feedback



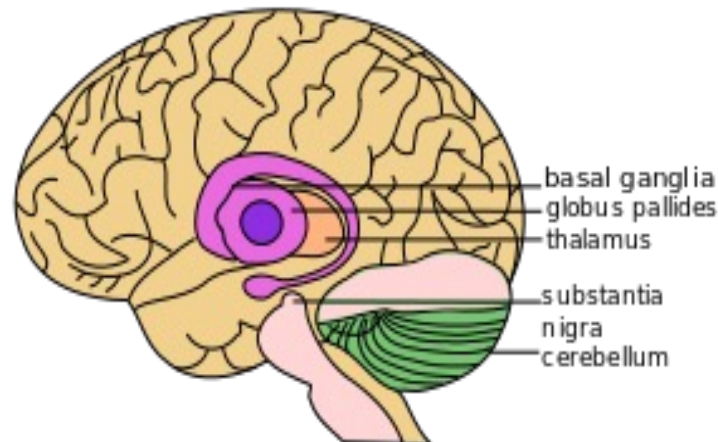
- **Control rhythm of CPG**
- **Tuning of motor movement**
- **Adaptive control**
- **Speed of locomotion**

Supraspinal



- **Balancing: Cerebellum**
- **Selection and control of gait postures**
- **Speed and direction of walk**
- **Visual guidance: visual directed limb positioning**
- **Control of reaching and grasping**
- **Anticipatory postural Adjustments**

Basal Ganglia and Related Structures of the Brain



Development of Walking

Neonatal Reflexes



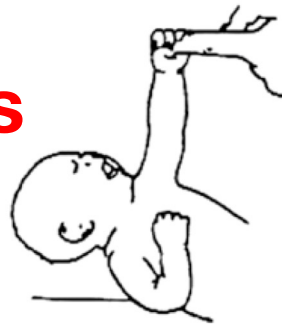
Startle/moro reflex



Tonic neck reflex



Stepping reflex



Grasp reflex



Grasp reflex

**Disappear @
different stages**

Reference for motor development

Navigation

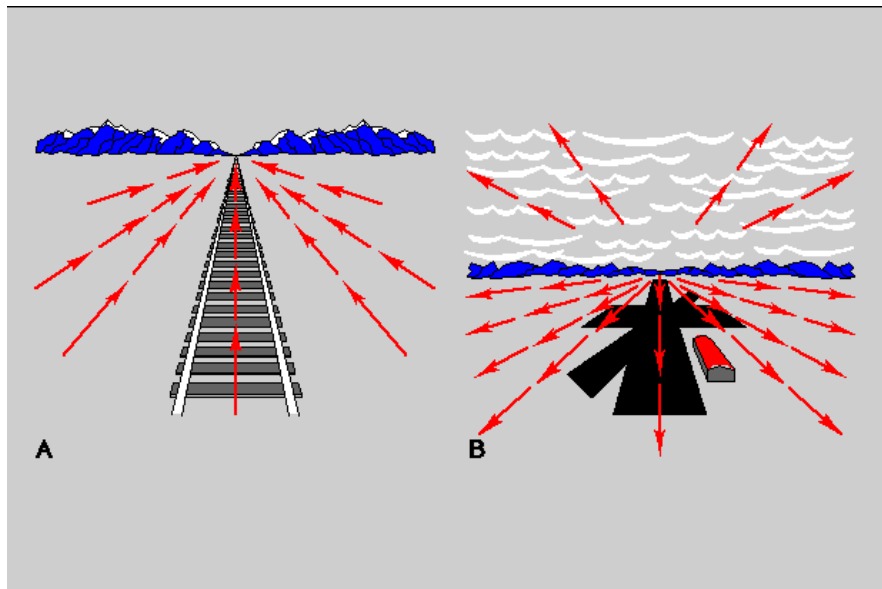
Navigation



<https://blog.directauto.com/road-trips-destinations/tips-safe-driving-mountains/>



<https://www.shutterstock.com/search/walk+through+valley>



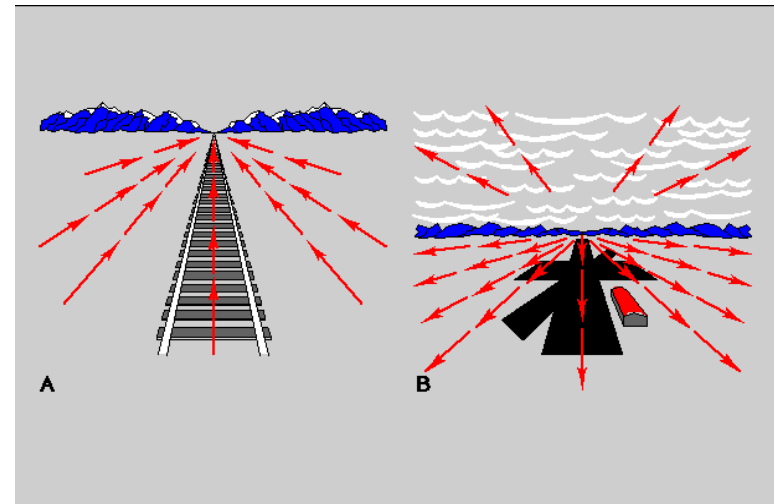
<https://web.northeastern.edu/cvl/vistars>

**Adaptive control of
whole body motion in the
external environment**

Optic Flow

Optic Flow Stimuli

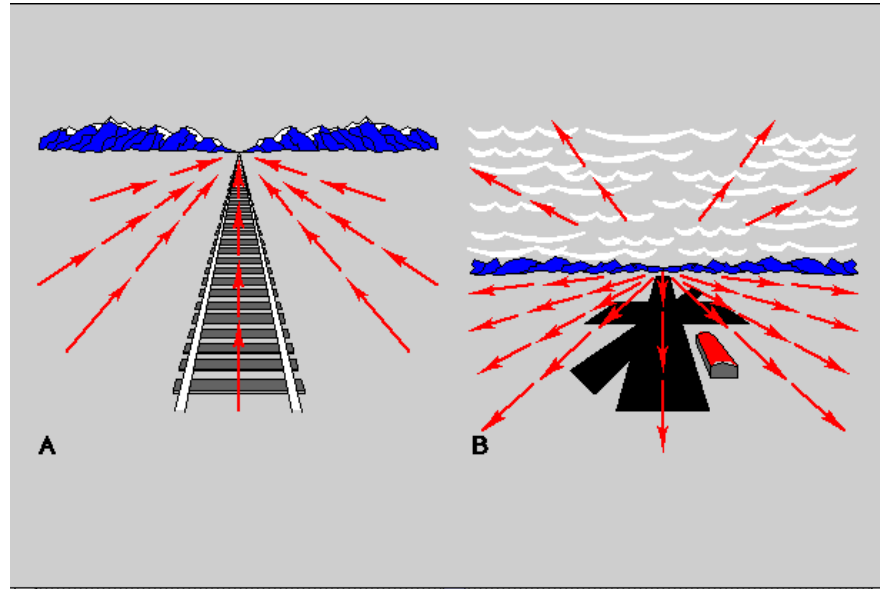
Form of stimuli generated when an observer navigates through its environment.



<https://web.northeastern.edu/cvl/vistars>

- **Gradient of speed:** Speed of objects is different at different places, e.g. Flow of objects is slower at the centre than at the periphery.
- **Focus of attention:** Absence of any flow at the centre of the screen or at the point of focus.

Navigation: Optic Flow

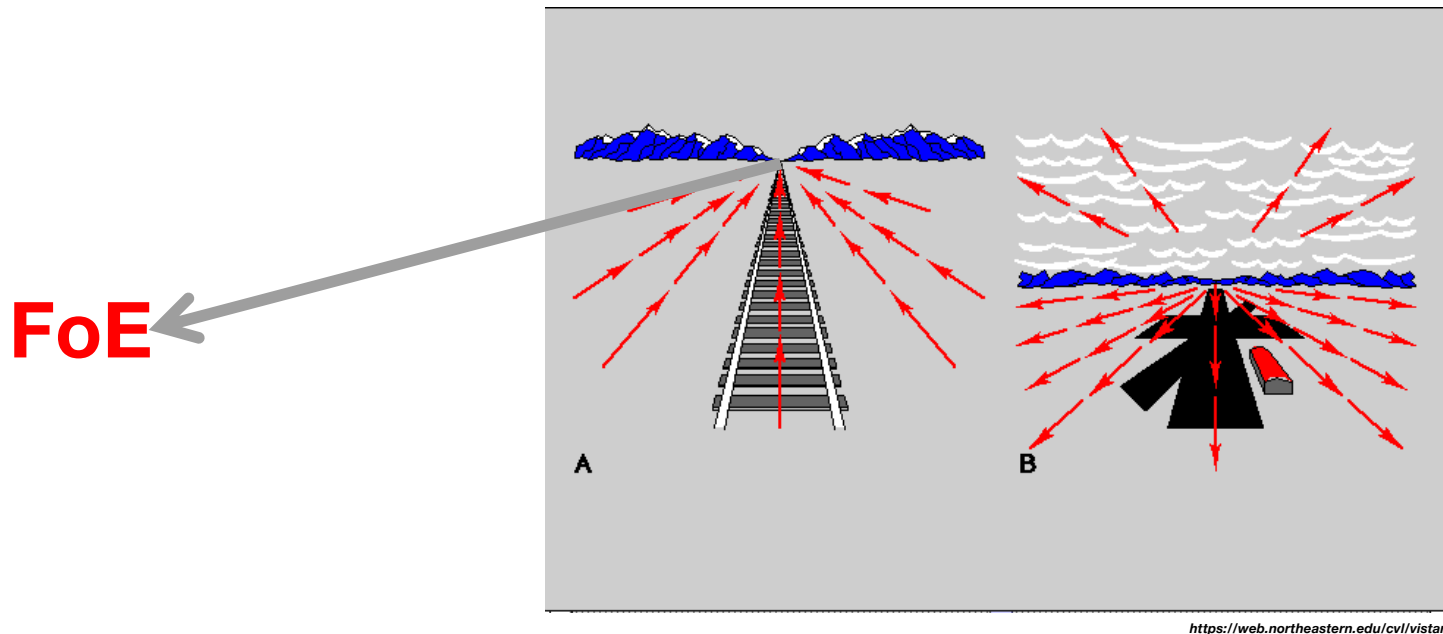


<https://web.northeastern.edu/cvl/vistars>

Navigation dependent upon:

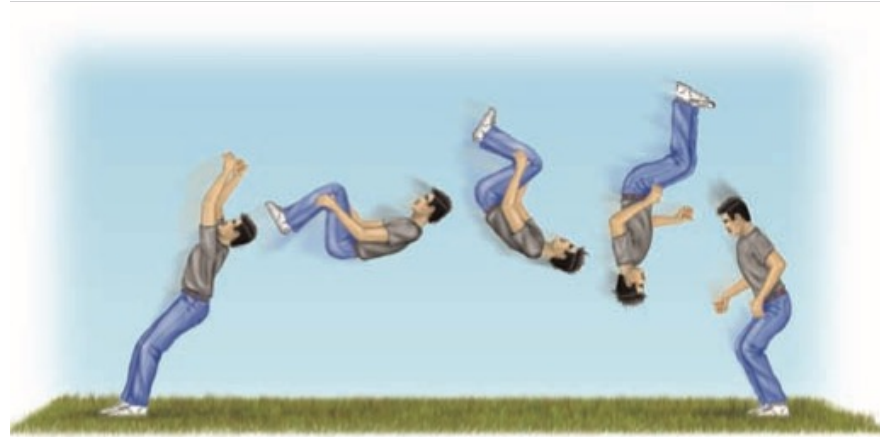
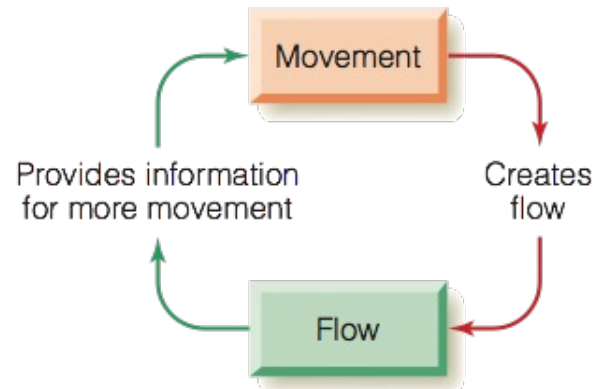
- **Perception**
- **Memory**

Optic Flow



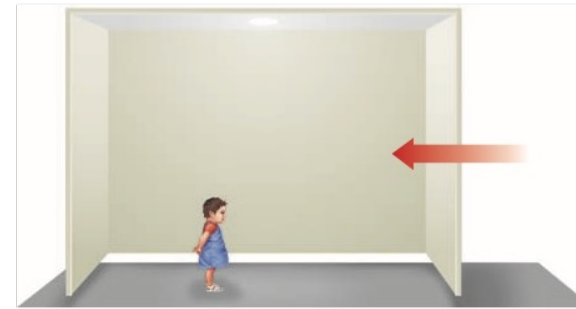
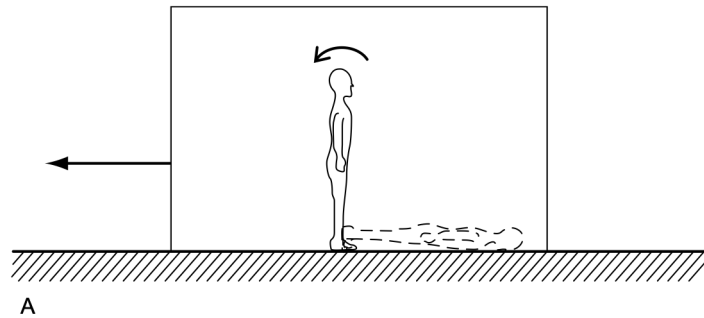
- Drives rapid eye movement to
 - Stabilize image in retina
 - Correspondence of images between two eyes
- Helps in stereovision/depth perception
- Help in self motion: relative position of objects w.r.t observer

Self- induced motion

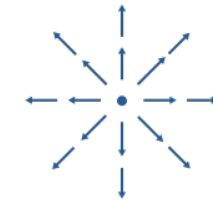


- One of the way that helps us navigate in our environment.
- Bardy et al, 1998 showed that trained gymnast could not execute somersaults with their eyes closed.
- But closed eyes did not effect performance of novice gymnast.
- These results indicates that we perform error correction during planning our actions.

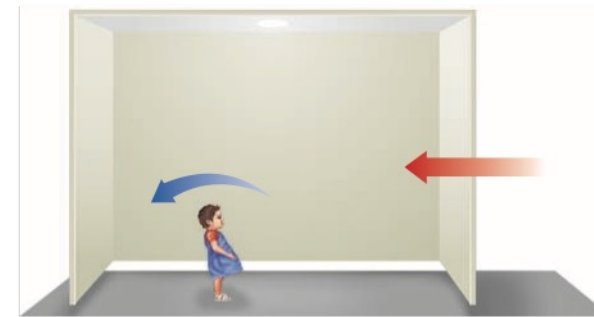
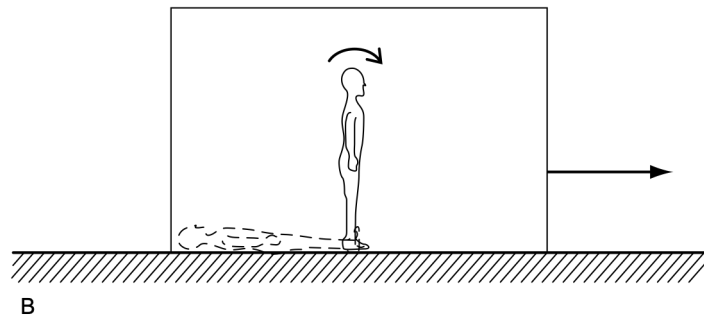
Visual Kinesthesia



(a) Room swings toward person.



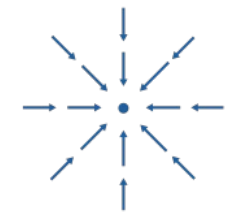
Flow when wall is moving toward person.



(b) Person sways back to compensate.



(c) When room swings away, person sways forward to compensate.



Flow when wall is moving away from person.

Lee & Aronson, 1974

Inputs from all sense organs

- Information from all the senses are collaborated and interpreted in order to execute an action.
- Important aspect of executing an action is to maintain balance.
- Traditionally balance of body is thought to be minted by inner ear canal.
- But its shown that balance is also maintained by vision (J.J. Gibson.)

Memory & Navigation

How to go Gate No 1 from R & D building

R& D Block → Park → Old Academic Building

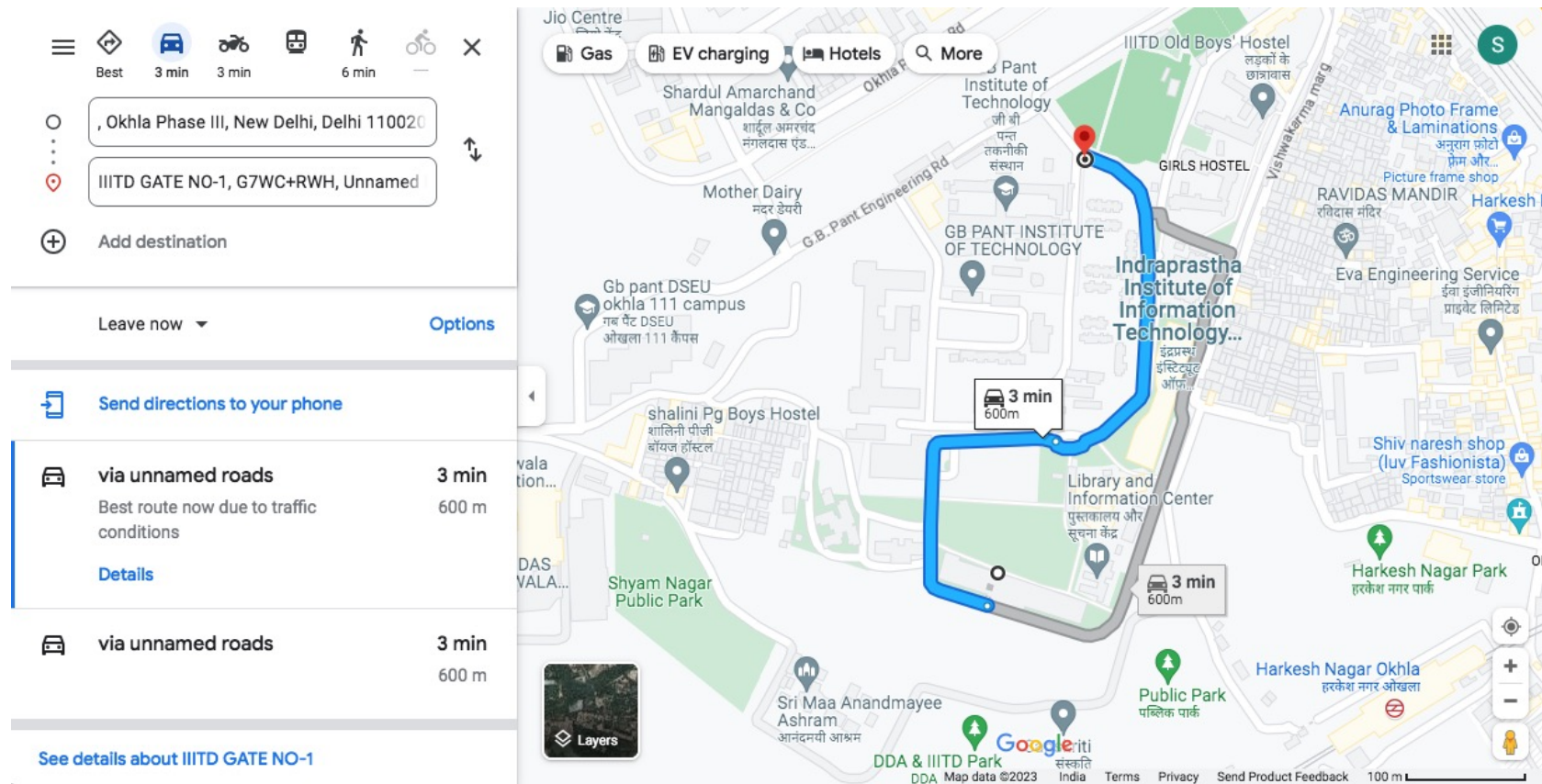
Route Maps

Lecture Block

Gate No. 1 ← Football ground ← Canteen

Memory & Navigation

How to go Gate No 1 from R & D building



Courtesy: Google Maps

Survey Maps

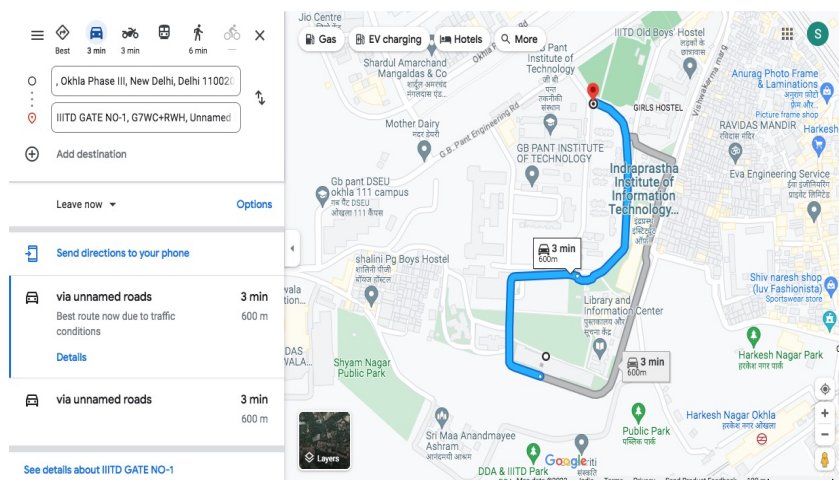
Route Vs Survey Maps

R& D Block → Park → Old Academic Building

Lecture Block

Route Maps

Gate No. 1 ← Football ground ← Canteen



Survey Maps

Courtesy: Google Maps